

PROJECT SYNOPSIS

Fintech Platform: AI-Powered Financial Analytics & Trading
Simulator

1. Executive Summary

This project is a state-of-the-art, AI-powered financial technology platform designed to provide institutional-level analytics, data processing, and predictive modeling to retail investors. The system seamlessly integrates real-time market data with advanced machine learning models to enable data-driven investment decisions, offering a comprehensive suite for trading simulation, portfolio management, and technical analysis.

2. Objectives

- To democratize access to advanced financial analytics and predictive machine learning models.
- To provide a robust paper-trading environment for testing and validating trading strategies without financial risk.
- To deliver real-time data integration alongside fundamental and technical analysis tools.
- To offer educational resources and a community platform to foster financial literacy and collaborative learning.

3. System Architecture

The platform utilizes a highly scalable microservices architecture encompassing three compute nodes and two stateful data nodes:

- **Frontend (UI Node):** A Next.js (React) application serving as a highly responsive user interface.
- **Backend (Core API Node):** A FastAPI (Python) service handling business logic, user management, and order execution.
- **ML Service (Analytics Node):** A dedicated FastAPI service for model training, inference, and heavy computational workflows.
- **Database:** PostgreSQL extended with TimescaleDB for optimized time-series market data storage.
- **Cache & Broker:** Redis for session management, fast caching, and asynchronous task queueing.

4. Key Modules & Features

4.1 Frontend Modules

- **Dashboard & Portfolio Management:** Centralized view of portfolio balance, daily P&L, top movers, and active alerts.
- **Technical & Fundamental Analysis:** Interactive charting with advanced indicators and comprehensive company health metrics.

- **ML Predictions & Risk Analysis:** Visualizes Value at Risk (VaR), portfolio beta, Sharpe ratios, and algorithmic strategy performance.
- **Backtesting & Trading Simulator:** Interactive simulation engine to test strategies and execute paper trades (Market, Limit, Stop orders).

4.2 Backend & ML Capabilities

- **Predictive Modeling:** LSTM/Transformer models for price forecasting with confidence intervals.
- **Sentiment Analysis:** NLP-driven evaluation of daily news and social feeds for bullish/bearish scoring.
- **Algorithmic Rule Engine:** Evaluates complex combinatorial logic for custom automated trading strategies.

5. Technology Stack

- **Frontend:** React.js, Next.js, Tailwind CSS, TypeScript
- **Backend API:** Python, FastAPI, SQLAlchemy, Pydantic
- **Machine Learning:** Python, PyTorch/TensorFlow, Scikit-learn, Pandas, NLTK/Transformers
- **Infrastructure:** Docker, Docker Compose, PostgreSQL (TimescaleDB), Redis

6. Future Scope

The platform is designed with future scalability in mind. Planned enhancements include:

- Integration of WebSocket channels for sub-millisecond real-time market data streaming.
- Live broker integrations (e.g., Alpaca, Interactive Brokers) to transition from paper trading to real capital execution.
- Kubernetes (K8s) orchestration for dynamic auto-scaling of ML computational nodes during peak market hours.